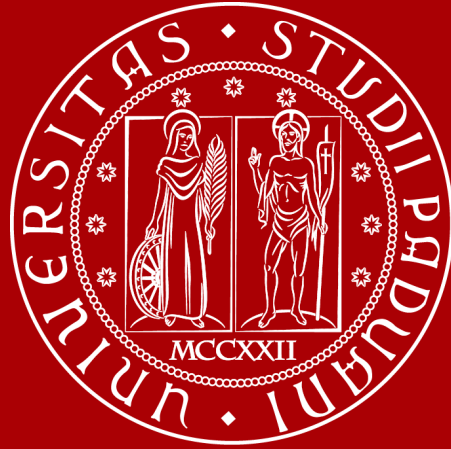




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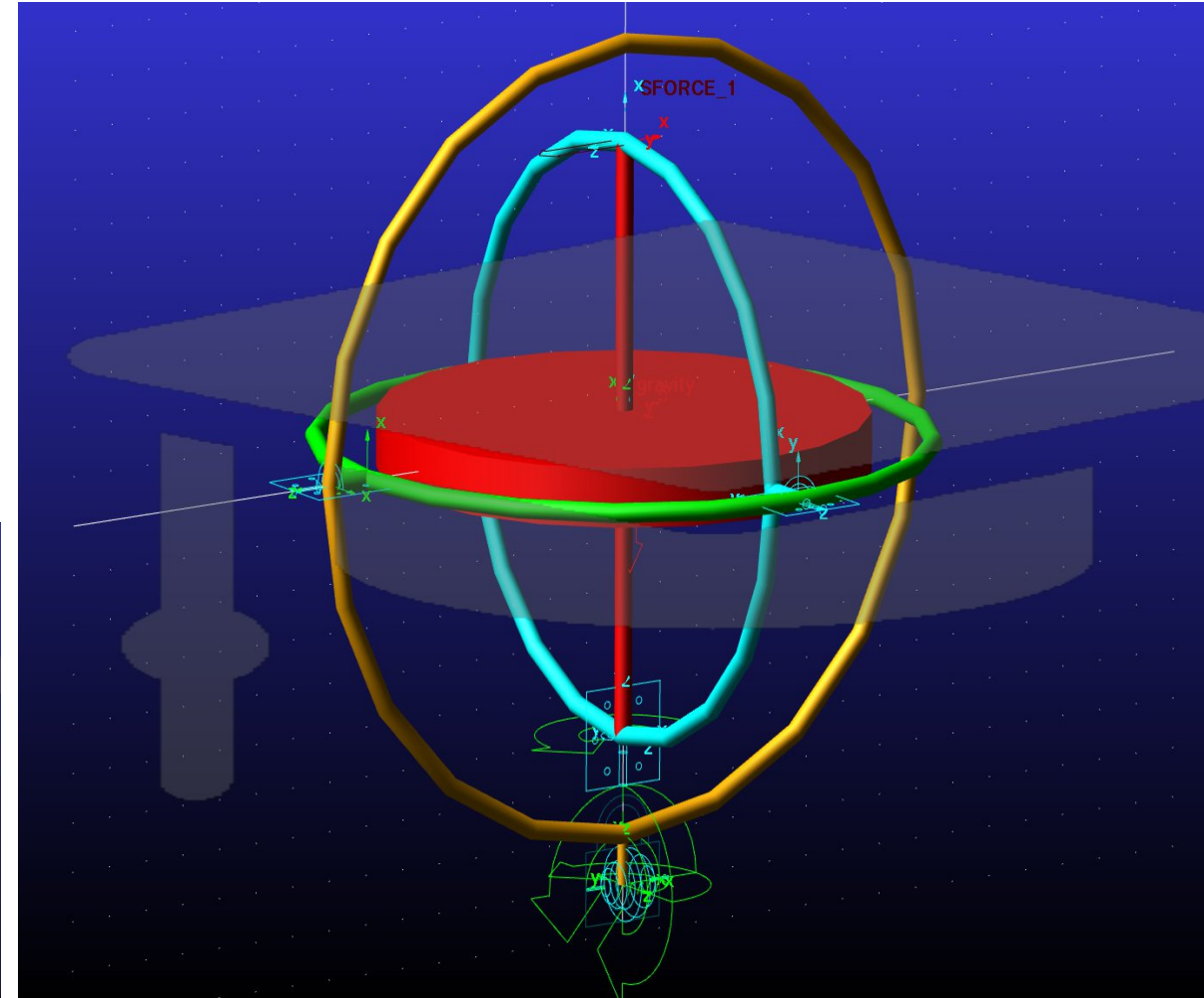
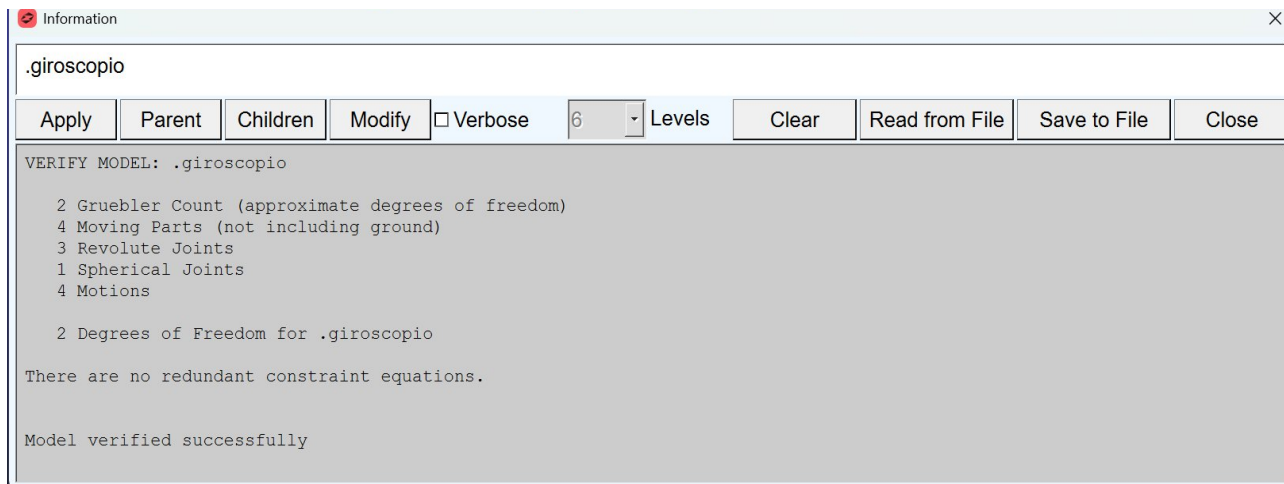
Stability of a spinning gyroscope

Modeling and simulation of mechanical systems project



MODEL DESCRIPTION

- Inertia wheel
- Inner gimbal
- Outer gimbal
- Frame





Objectives

The aim of this study is to verify the two properties of a gyroscope:

- 1) Rigidity in space → move the frame
- 2) Precession → create a force

1) Rigidity in space



Fixed orientation regardless of any movement of its mounting

“Gyroscope rigidity in space is the property of a spinning gyroscope to maintain a fixed orientation regardless of any movement of its mounting”

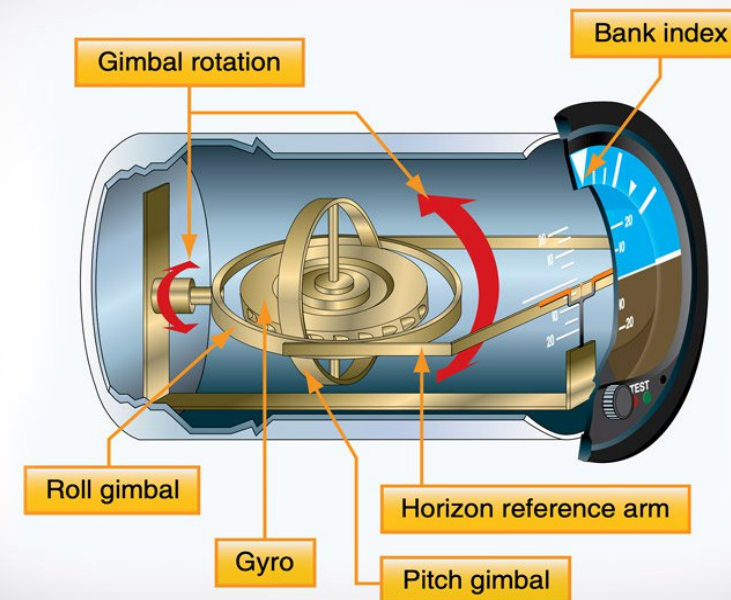
A gyroscope will remain rigid in space



The airplane “moves around the gyro”



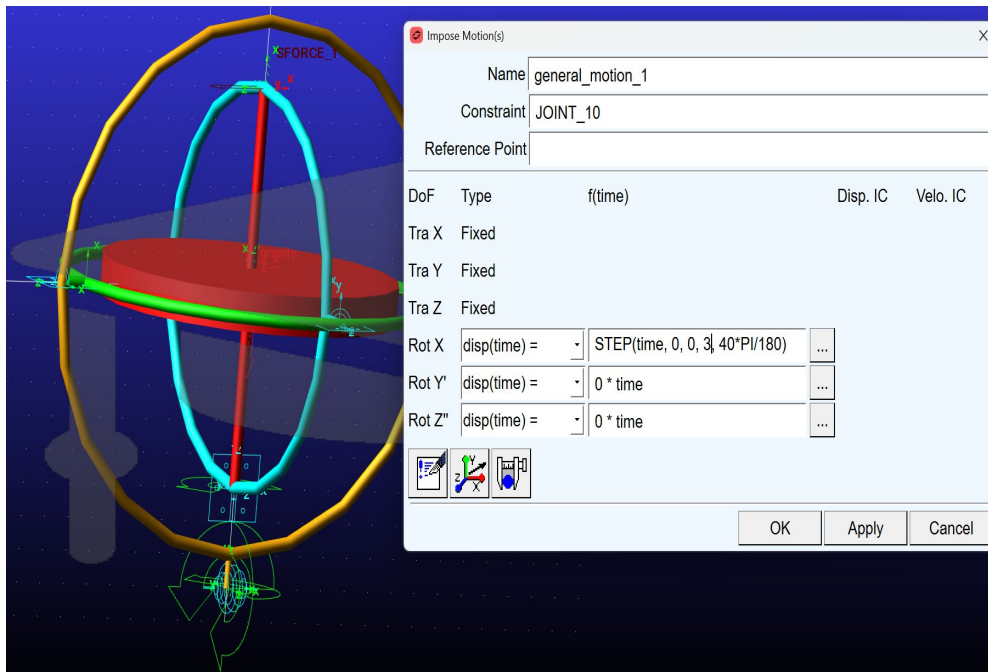
Attitude Indicator Diagram



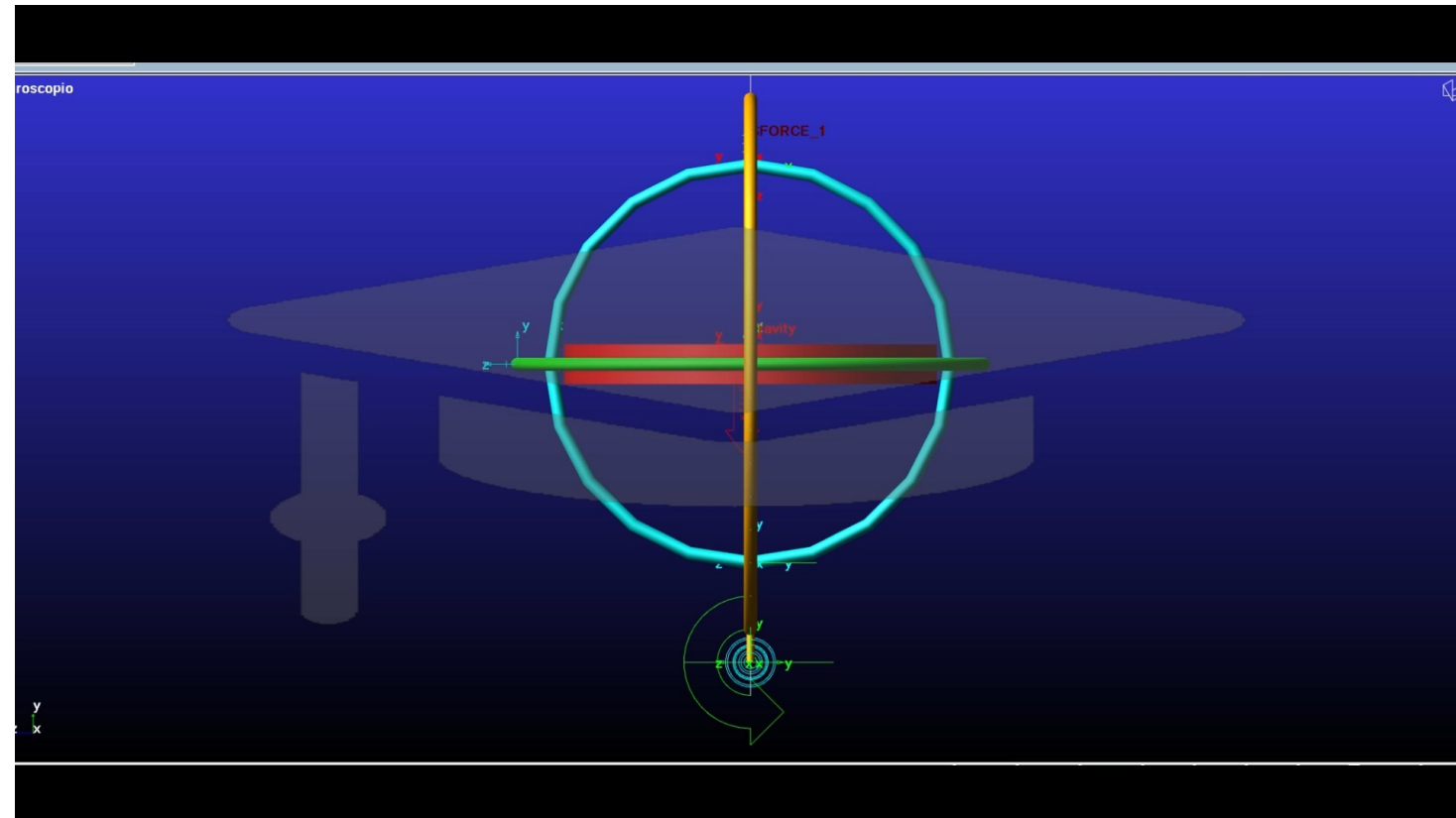


One axis rotation

Imposed motion on the frame:



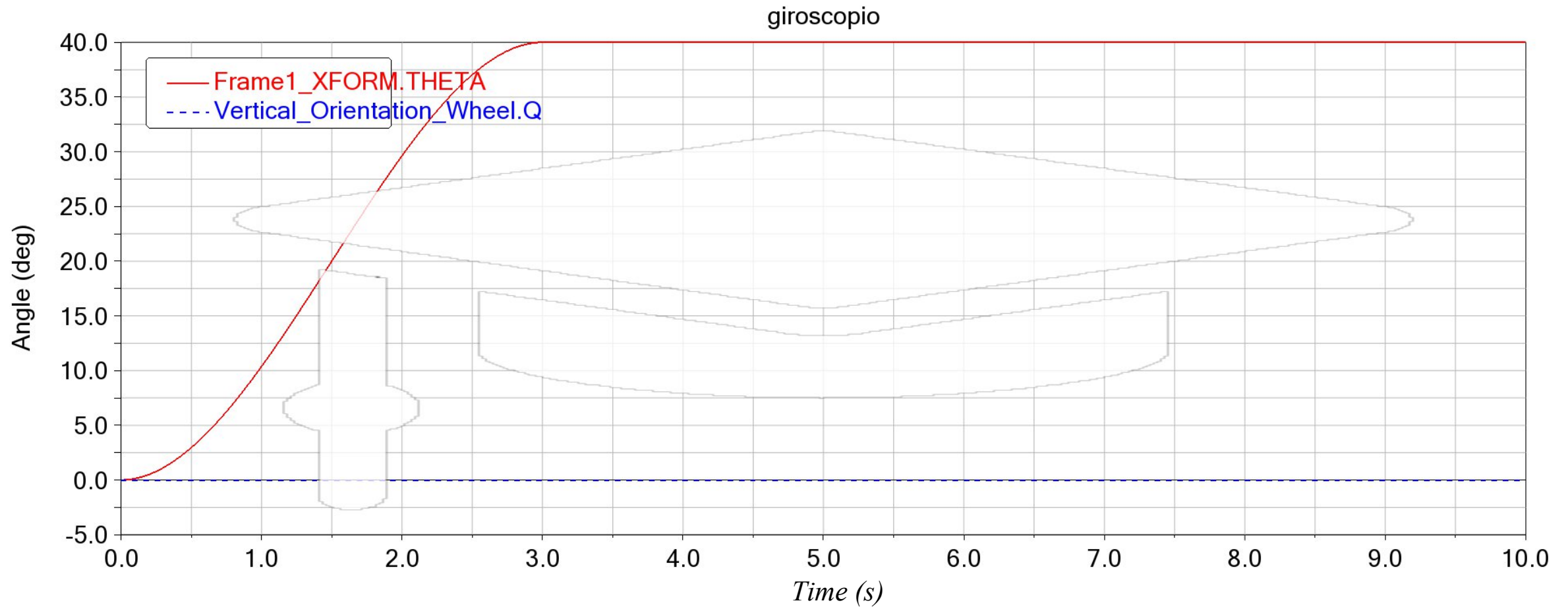
Video reference:





Result:

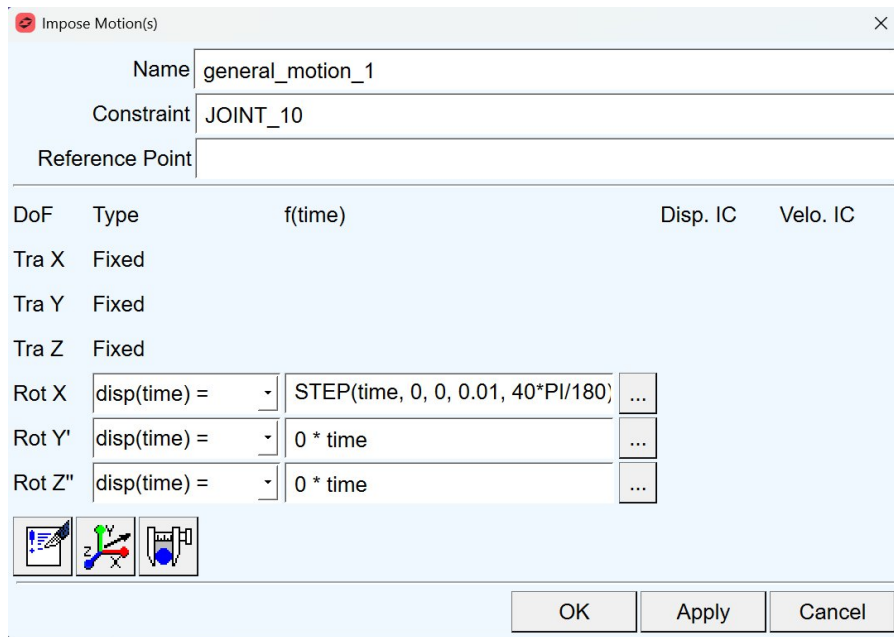
Rotational speed fixed at 2000 rad/s





Istant rotation of the frame? No problem!

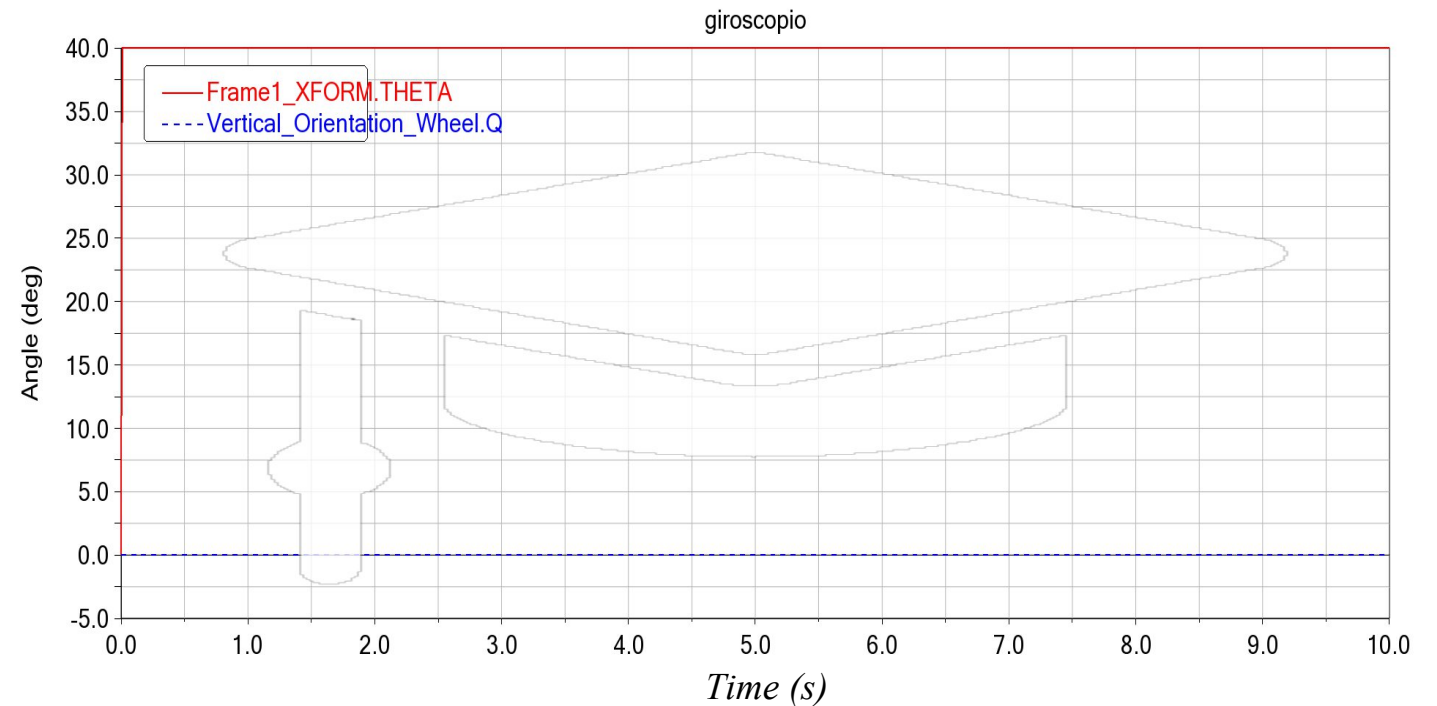
STEP(time, t_start, beginning_value, t_end, end_value)



Impose Motion(s) dialog box configuration:

- Name: general_motion_1
- Constraint: JOINT_10
- Reference Point: (empty)
- DoF Type f(time) Disp. IC Velo. IC
- Tra X Fixed
- Tra Y Fixed
- Tra Z Fixed
- Rot X disp(time) = STEP(time, 0, 0, 0.01, 40*PI/180)
- Rot Y' disp(time) = 0 * time
- Rot Z'' disp(time) = 0 * time

Results



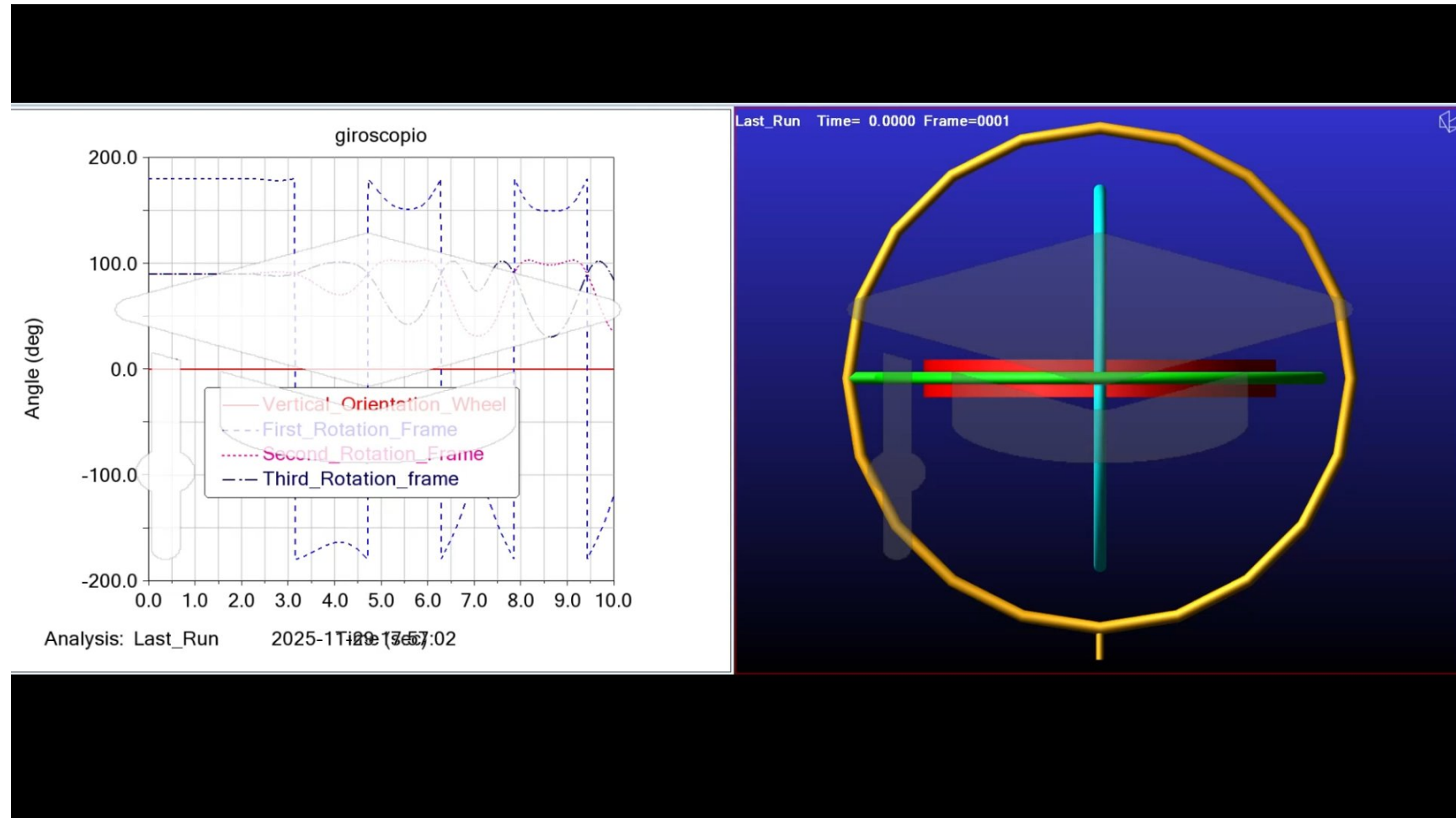
Rotational speed fixed at 2000 rad/s



Three axis rotations of the frame

Rotations imposed:

Rot X	disp(time) =	step(time, 2,0,7,1)*((45*PI/180)*sin(2*time))
Rot Y'	disp(time) =	step(time, 2,0,7,1)*((45*PI/180)*sin(2*time))
Rot Z''	disp(time) =	step(time, 2,0,7,1)*((45*PI/180)*sin(2*time))



Rotational speed fixed at 2000 rad/s

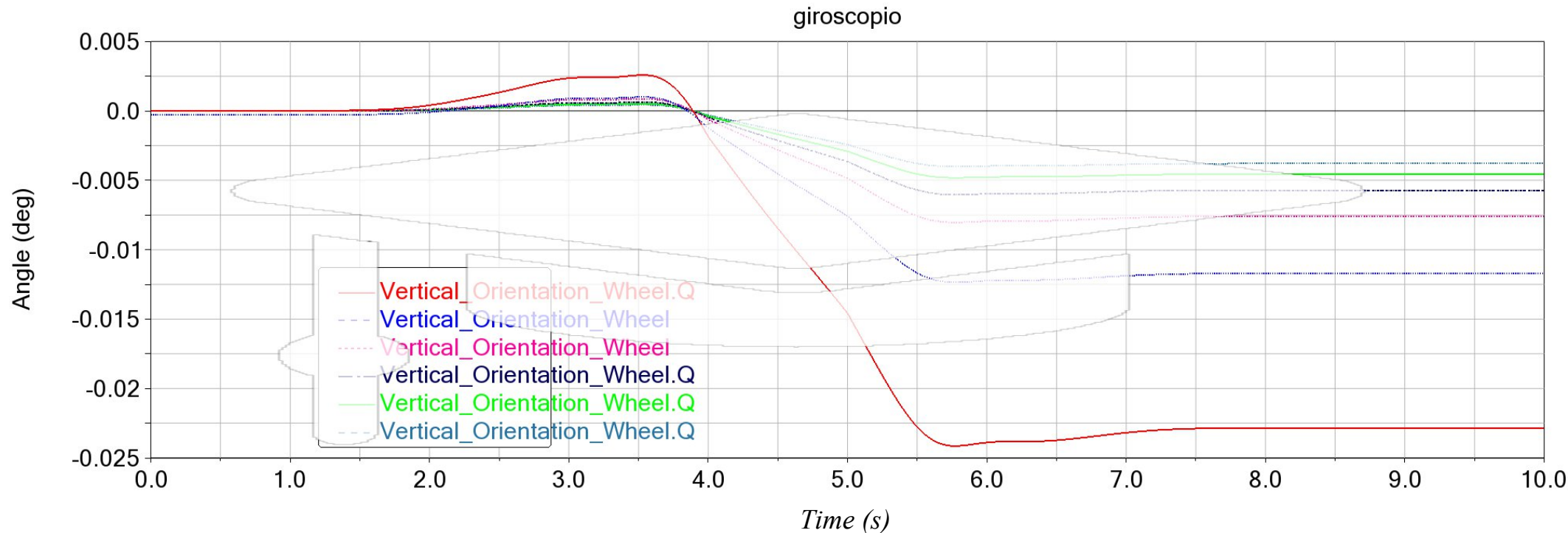


Vertical orientation isn't perfectly 0!

Parametric study of the vertical orientation of the wheel changing the rotational speed

Fixed moment of inertia about x-axis : 4.5191899537E+06

- 1000 rad/s
- 2000 rad/s
- 3000 rad/s
- 4000 rad/s
- 5000 rad/s
- 6000 rad/s



Rotation applied:

Rot X	disp(time) =	step(time, 1,0,4,1)*step(time,5,1,8,0)*((45*PI/180)*sin(1*time))	...
Rot Y'	disp(time) =	step(time, 1,0,4,1)*step(time,5,1,8,0)*((45*PI/180)*sin(1*time))	...
Rot Z''	disp(time) =	step(time, 1,0,4,1)*step(time,5,1,8,0)*((45*PI/180)*sin(1*time))	...



Vertical orientation isn't perfectly 0!

Parametric study of the Vertical orientation of the wheel changing the width so changing its inertia about the x-axis

Fixed rotational speed: 3000 rad/s

Wheel width: 60mm

Moment of inertia about x-axis: $4.52\text{E}+06 \text{ Kg}\cdot\text{mm}^2$

Wheel width: 80mm

Moment of inertia about x-axis: $6.03\text{E}+06 \text{ Kg}\cdot\text{mm}^2$

Wheel width: 100mm

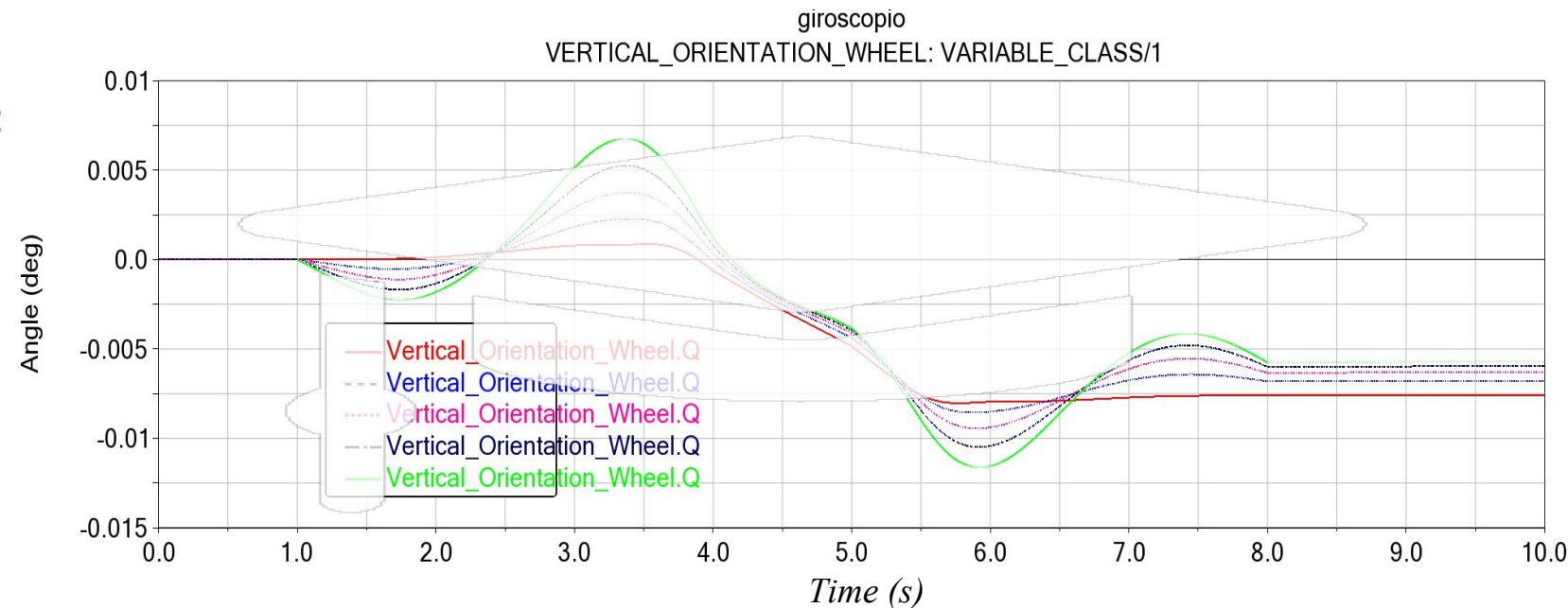
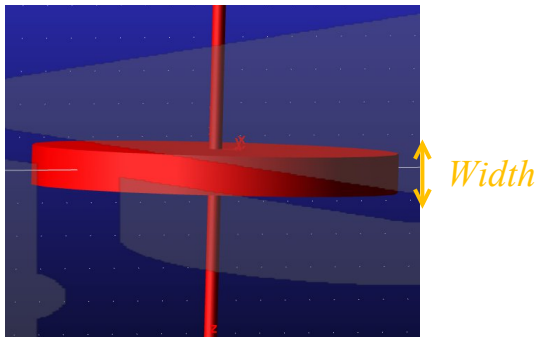
Moment of inertia about x-axis: $7.53\text{E}+06 \text{ Kg}\cdot\text{mm}^2$

Wheel width: 120mm

Moment of inertia about x-axis: $9.04\text{E}+06 \text{ Kg}\cdot\text{mm}^2$

Wheel width: 140mm

Moment of inertia about x-axis: $1.05\text{E}+07 \text{ Kg}\cdot\text{mm}^2$



Rotation applied:

Rot X	disp(time) =	step(time, 1,0,4,1)*step(time,5,1,8,0)*((45*PI/180)*sin(1*time))	...
Rot Y'	disp(time) =	step(time, 1,0,4,1)*step(time,5,1,8,0)*((45*PI/180)*sin(1*time))	...
Rot Z''	disp(time) =	step(time, 1,0,4,1)*step(time,5,1,8,0)*((45*PI/180)*sin(1*time))	...

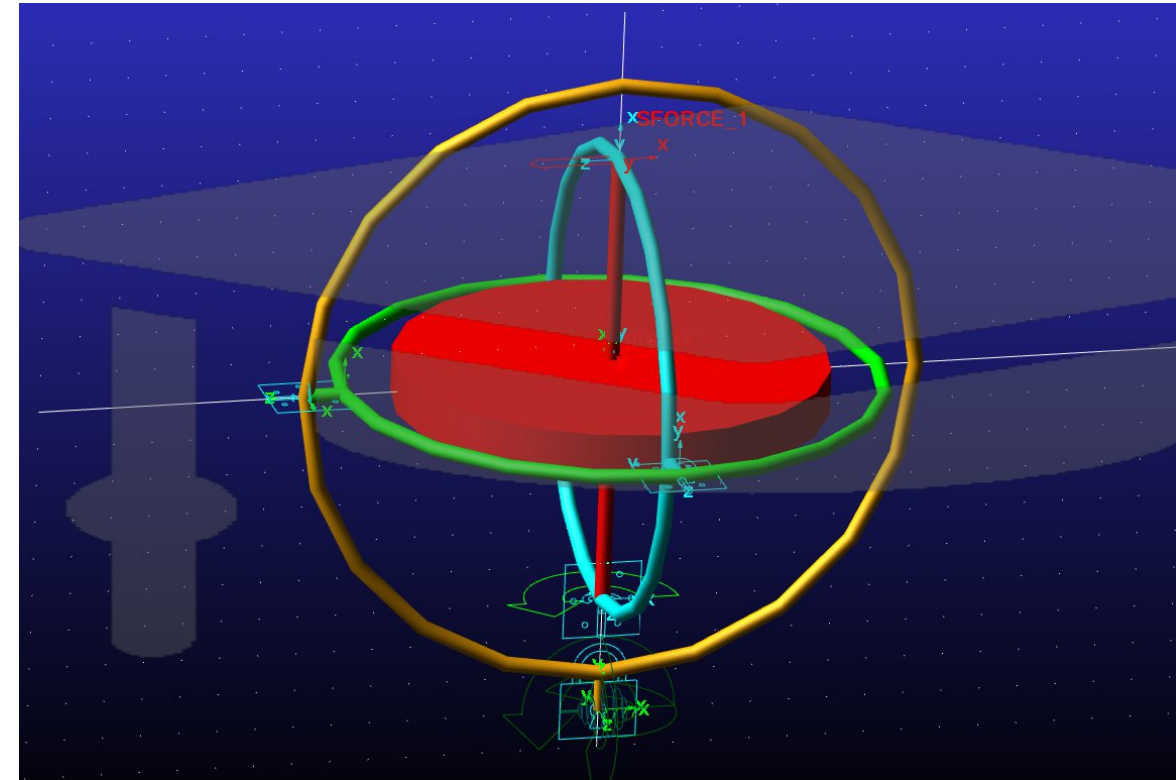


2) Precession

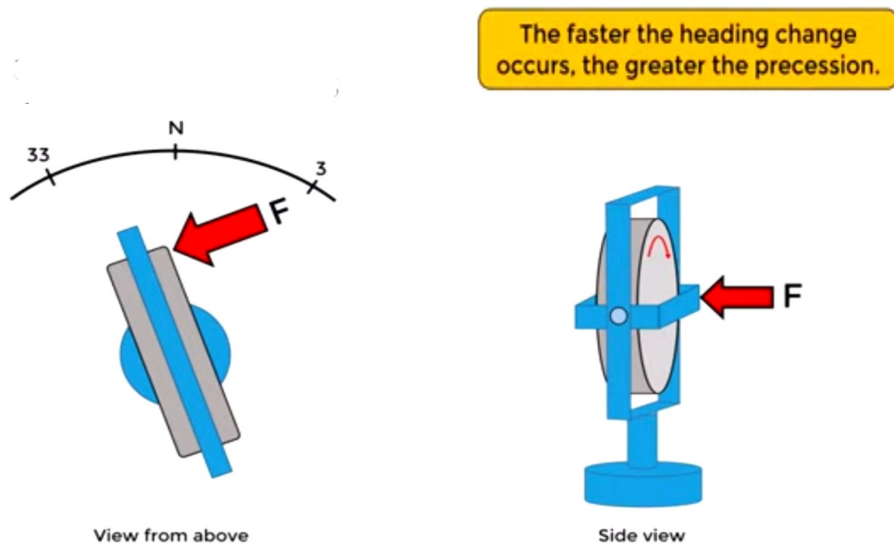
“Precession is the slow, continuous change in the direction of a gyroscope's spin axis when a constant torque is applied.

Instead of tipping over in the direction of the force, a spinning gyroscope rotates around a third axis, perpendicular to both the angular momentum and the applied torque”

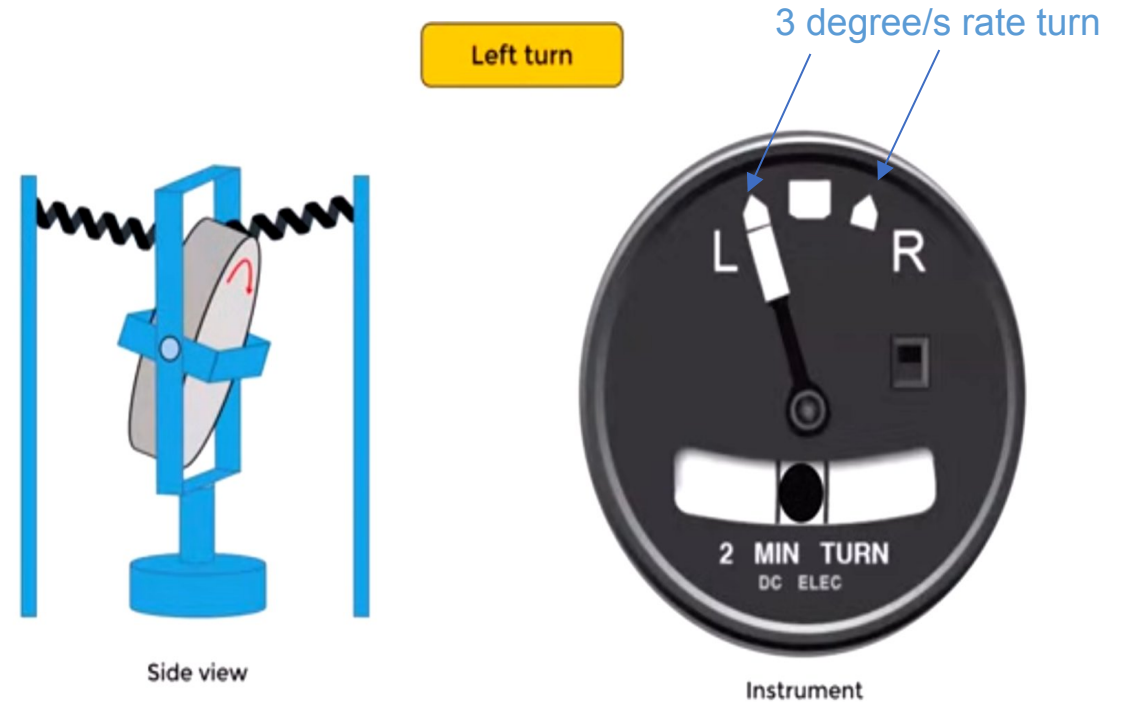
Created a torque applying a force showed in the image below



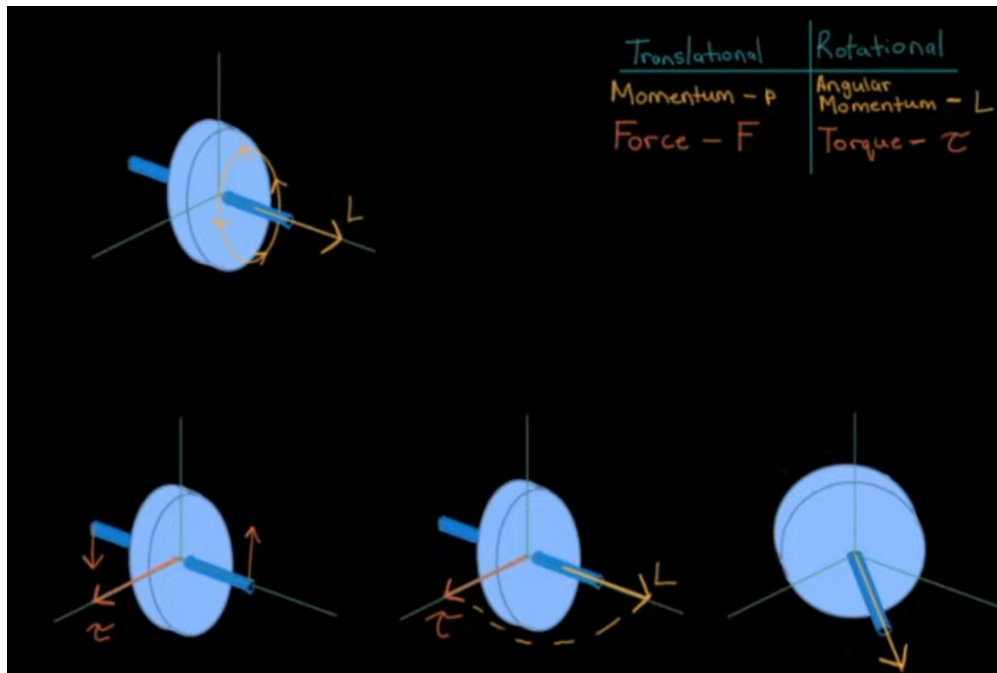
Precession applications



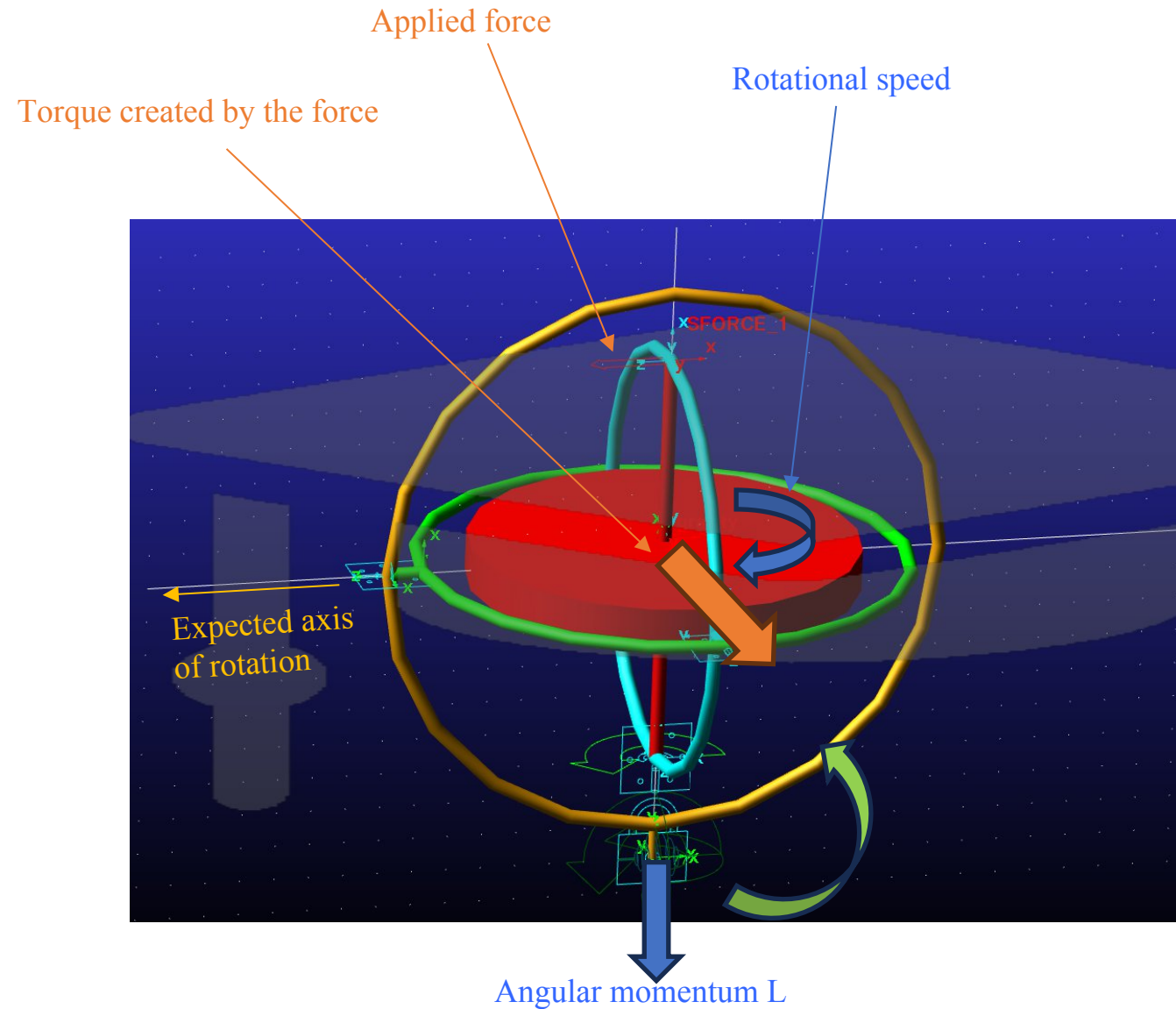
The movement of the gyro is directly transmitted to the needle on the instrument.



Costant force applied



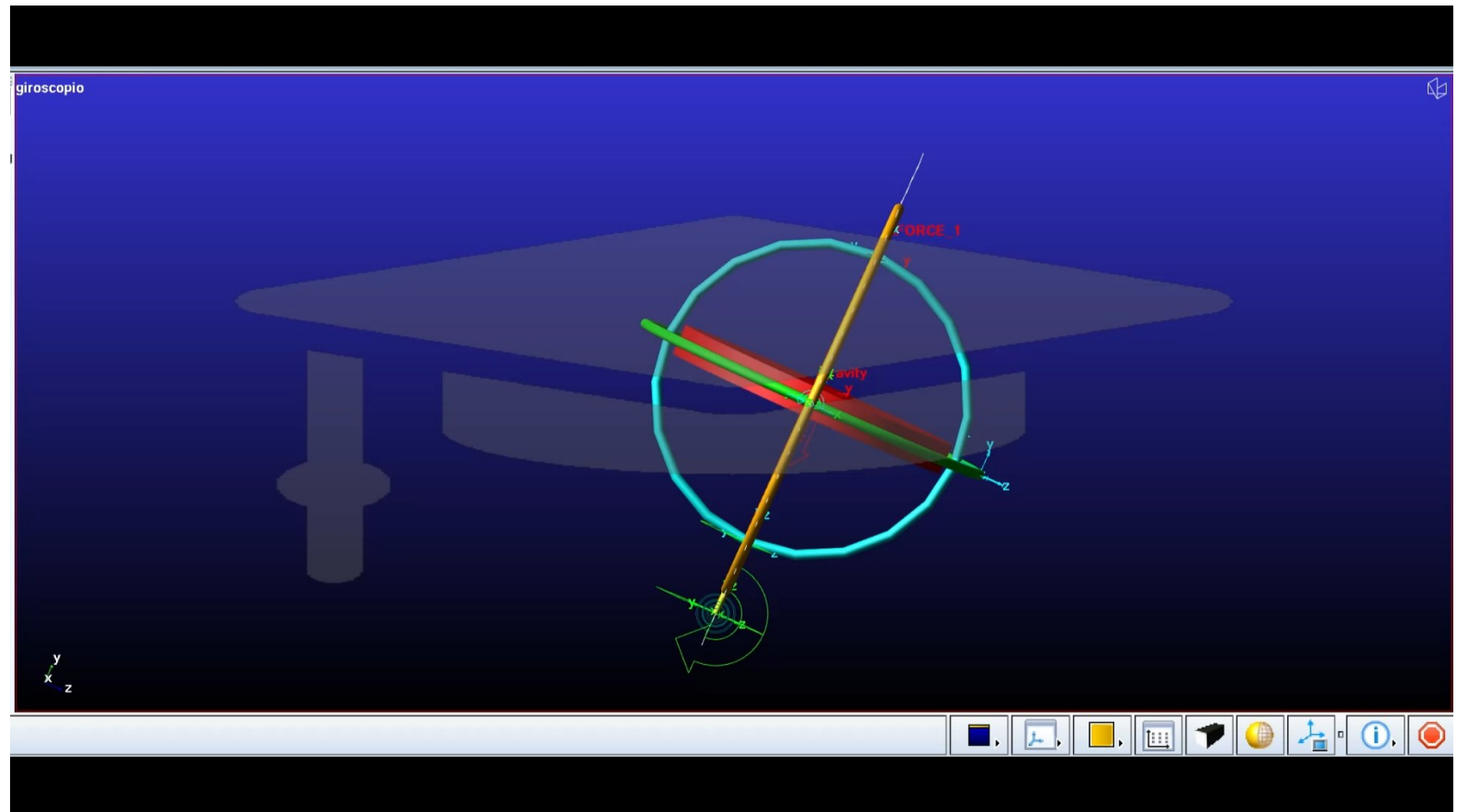
The angular momentum follows the direction of the torque so we can expect a rotation in the direction of the green arrow





Costant force applied

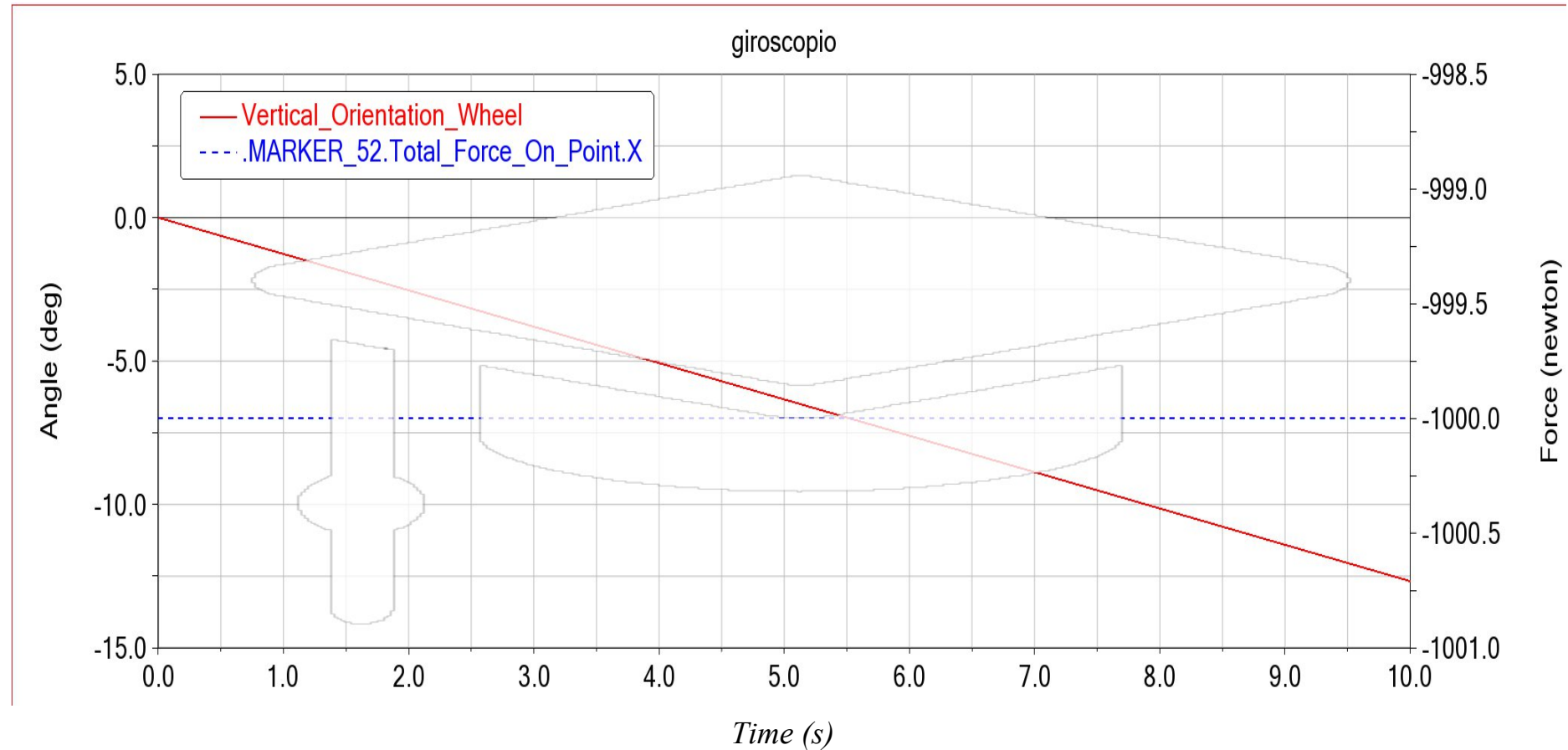
The video confirms the rotation predicted by the green arrow





Costant force applied

We can verify that the gyroscope changes its orientation constantly when a fixed force (1000N) is applied

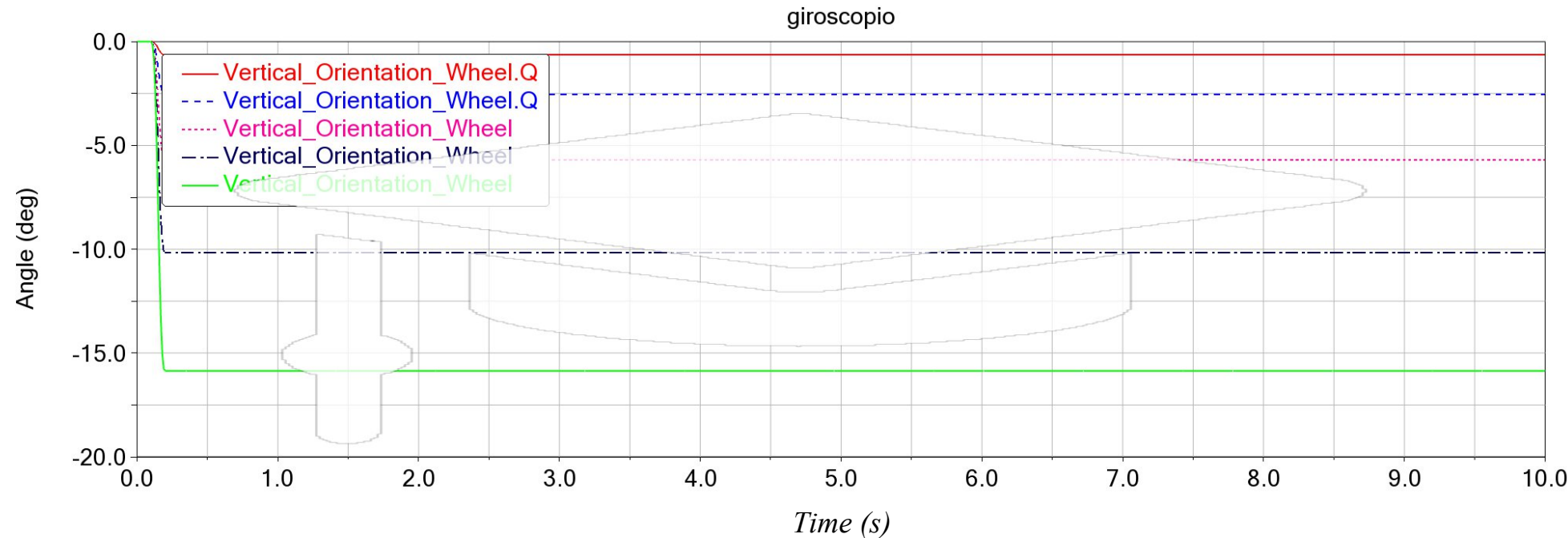




Applying an impulse disturbance

Parametric study of the vertical orientation with different values of the impulse force with inertia about x-axis ($4.519\text{E}+06 \text{ Kg}\cdot\text{mm}^2$) and rotation speed (2000 rad/s) fixed

- 100 Newton
- 200 Newton
- 300 Newton
- 400 Newton
- 500 Newton



`step(time, 0.1,0,0.15,100)*step(time,0.15,100,0.2,0)`: this function simulates an impulse disturbance applied upon inertia wheel that has a width of 0.1 second and an amplitude of 100 N



Applying an impulse disturbance

Parametric study of the vertical orientation with different values of the inertia about x-axis with speed (2000 rad/s) and impulse force (400 N) fixed

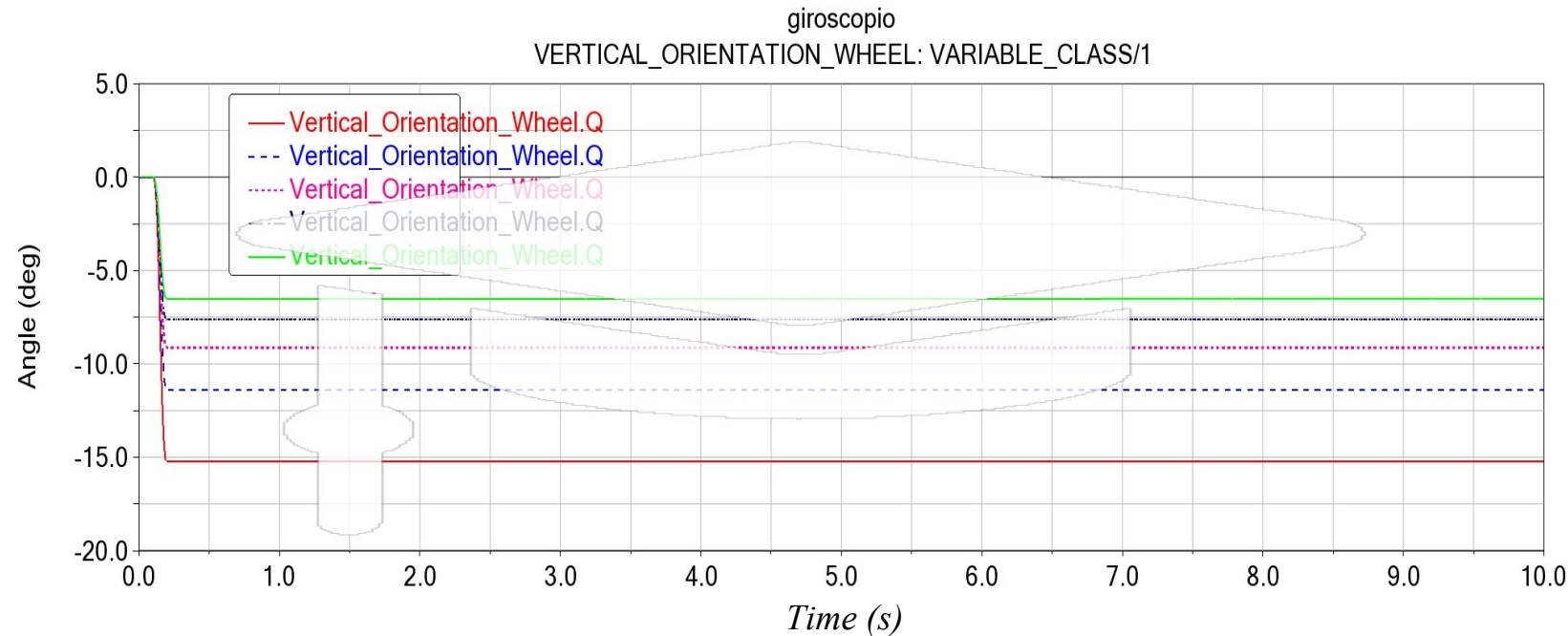
Wheel width: 60mm
Moment of inertia about x-axis: $4.52\text{E}+06 \text{ Kg}\cdot\text{mm}^2$

Wheel width: 80mm
Moment of inertia about x-axis: $6.03\text{E}+06 \text{ Kg}\cdot\text{mm}^2$

Wheel width: 100mm
Moment of inertia about x-axis: $7.53\text{E}+06 \text{ Kg}\cdot\text{mm}^2$

Wheel width: 120mm
Moment of inertia about x-axis: $9.04\text{E}+06 \text{ Kg}\cdot\text{mm}^2$

Wheel width: 140mm
Moment of inertia about x-axis: $1.05\text{E}+07 \text{ Kg}\cdot\text{mm}^2$



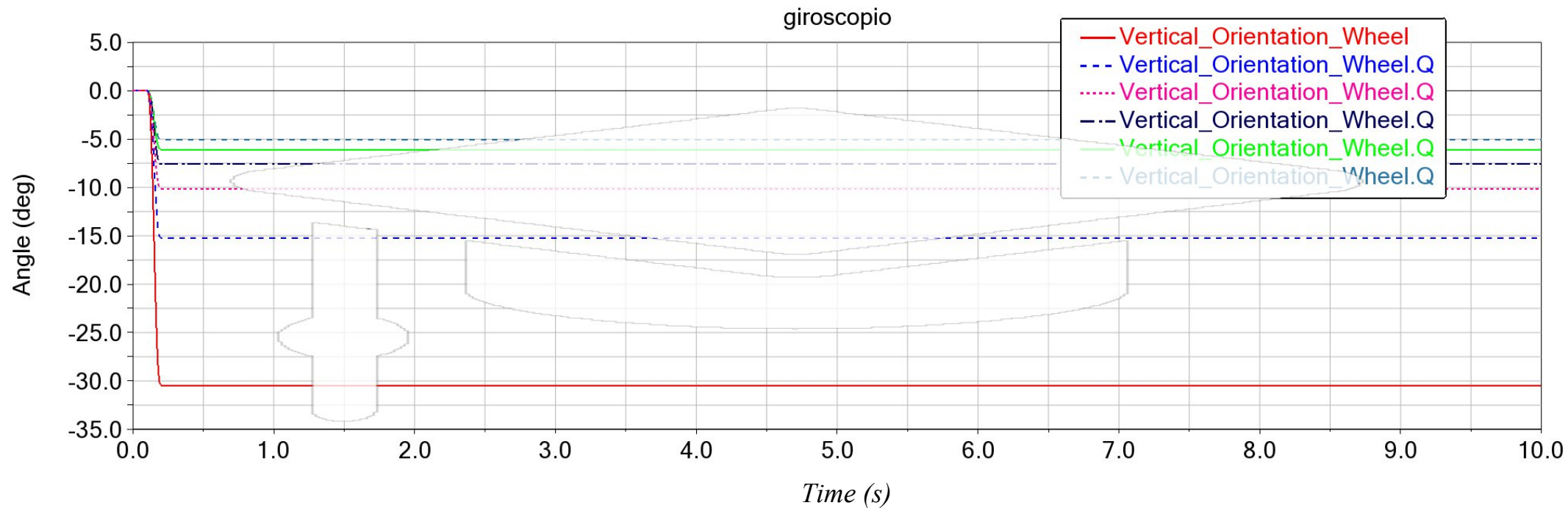


Applying an impulse disturbance

Parametric study of the vertical orientation of the wheel changing the rotational speed while applying an impulse force of 400N

Fixed moment of inertia about x-axis : 4.5191899537E+06

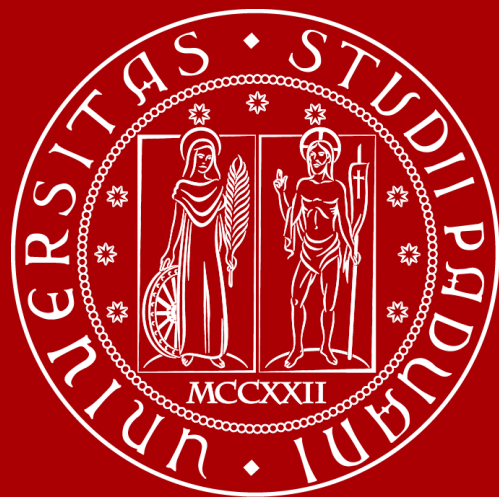
- 1000 rad/s
- 2000 rad/s
- 3000 rad/s
- 4000 rad/s
- 5000 rad/s
- 6000 rad/s





Conclusion

After the conducted parametric studies, it can be confirmed that the faster a gyroscope spins and the greater its inertia, the more stable and reliable its measurements become.



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